BioDINO-Edge: Biologically-Grounded Multimodal Few-Shot Learning for Ultra-Fast Crop Pathology on Resource-Constrained Micro-Scouts

Anonymous Authors

***Abstract*—Rapid and accurate foliar disease diagnosis is vital for global food security, yet existing deep learning paradigms fall short of practical agricultural deployment. Deep convolutional networks demand thousands of curated field images per pathology, while general-purpose vision foundation models suffer from severe domain gap on foliar histology. Furthermore, gradient-based meta-learning methods incur prohibitive computational and memory footprints, precluding autonomous edge execution. To overcome these limitations, we introduce BioDINO-Edge, an adaptive dual-foundation framework uniting self-supervised visual patch tokens (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher Linear Discriminant Analysis (Fisher-LDA) with closed-form Soft-Anchored Transductive Self-Calibration, BioDINO-Edge establishes an optimal foliar manifold that suppresses shared vegetative background variance by over $31\times$ and eliminates small-sample covariance singularity ($N \ll d$) via analytical Ledoit–Wolf shrinkage without backpropagation. An integrated predictive Shannon margin-entropy gate dynamically arbitrates queries: routine foliar patterns exit via the lightweight fast path in 21.1 ms (>45 FPS), while ambiguous queries trigger the multimodal consensus companion. Across the 11-class PlantVillage benchmark under strict 5-shot constraints, BioDINO-Edge establishes a new state of the art with 98.72% peak accuracy, 98.41% peak F1-score, and a 30-episode mean of 97.37% $\pm$ 0.73%, outperforming CLIP+LoRA by 10.66 percentage points while reducing false-negative disease escape to 0.604%. On the uncontrolled in-the-wild PlantDoc dataset, it attains 83.61% 5-way 5-shot accuracy (+13.00% over prior SOTA) and 56.35% in 27-way field stress testing. Compiling into a standalone 22.73 MB native Android APK operating at sub-22 ms latency alongside an on-demand 83.57 MB companion, BioDINO-Edge delivers a field-ready, mathematically rigorous diagnostic capability for autonomous agricultural scouting.**

***Index Terms*—Few-Shot Learning, Precision Agriculture, Plant Pathology, Multimodal Foundation Models, Fisher Discriminant Analysis, Edge AI, Transductive Inference.**

I. Introduction

Plant pests and diseases destroy an estimated 20%–40% of crops annually [1]. Traditional diagnosis of plant pathogens heavily relies on visual leaf characteristics such as color, shape, and lesions [2]. In field practice, visual inspection is slow, depends on trained scorers, and becomes ambiguous when different pathogens produce similar lesions [3]; faster and more reproducible identification is therefore needed.

Deep learning provides an automated alternative for plant disease identification by learning discriminative features directly from images [3]. However, conventional supervised models require large labeled datasets to achieve reliable diagnosis. Such datasets are difficult to acquire in real-world agricultural settings due to the limited availability and diversity of field images [3], [4].

FSL has become a common workaround: instead of collecting thousands of labeled images, it trains a model to recognize new disease classes from a handful of support examples. Recent research has attempted to apply FSL using cross-domain few-shot learning models—such as CDFSL-BDC [5] and MAFDE-DN4 [6]—as well as few-shot metric domain adaptation (FMDA) [7] and visual-semantic fusion (TSMF) [8], alongside recent edge-federated approaches [9]. However, these approaches exhibit several practical limitations:

- **Domain Shift & Target-Domain Dependency:** Cross-domain models (such as CDFSL-BDC [5] and MAFDE-DN4 [6]) perform poorly under cross-domain evaluation due to complex backgrounds and multi-scale lesions, while metric adaptation methods (such as FMDA [7]) still require labeled samples from the target domain.
- **Semantic Misalignment:** Multimodal models relying on LLM text descriptions (e.g., GPT-4o in TSMF [8]) fail to generalize well to real-world examples, suffering from semantic inaccuracies and a lack of fine-grained alignment.
- **Deployment Inefficiencies:** Federated and edge models (e.g., [9]) lack sufficient testing using real-world data streams, making them inefficient for real-time edge deployment.

We introduce BioDINO-Edge, a dual-foundation pipeline tailored to cross-domain few-shot plant pathology under edge-hardware constraints. It combines visual patch tokens from DINOv2 [10] with clean botanical domain prompts from BioCLIP [11], eliminating the need to fine-tune foundation models to detect subtle necrotic lesions. Using closed-form Fisher-LDA [12] and Ledoit–Wolf shrinkage [13], our model completely removes backpropagation during adaptation, enabling rapid on-device calibration in milliseconds.

The key contributions of this work are as follows:

1) **Dual-Foundation Foliar Representation:** Uniting self-supervised visual patch tokens from DINOv2 with purified botanical domain prompts from BioCLIP to resolve subtle necrotic lesions without foundation-model fine-tuning.
2) **Closed-Form Analytical Adaptation:** Integrating Fisher-LDA with Ledoit–Wolf covariance shrinkage and soft-anchored transductive recalibration to overcome high-dimensionality issues and 5-shot centroid shifts within

25 milliseconds without backpropagation.

3) **Adaptive Two-Tier Edge Deployment:** Routing queries through a Shannon [14] margin-entropy gate to execute an ultra-lightweight 22.73 MB standalone APK alongside an 83.6 MB on-demand module within a strict 110 MB memory footprint.
4) **Rigorous Cross-Domain Validation:** Under the standard 5-way 5-shot protocol, BioDINO-Edge outperforms the multimodal VSF baseline by 13.00 percentage points on PlantDoc and by 6.86 percentage points on PlantVillage, while remaining within edge-device memory budgets.

II. Proposed BioDINO-Edge Architecture

Problem Formulation and Mathematical Setting. We formulate cross-domain few-shot foliar pathology classification as follows. Let $\mathcal{X} \subset \mathbb{R}^{3\times H\times W}$ denote the RGB image domain, and let $\mathcal{Y} = \{1, \ldots, C\}$ denote the label space of C novel phytopathological categories. During adaptation, the learner receives a small support set $\mathcal{S} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_s}$ containing exactly K labeled leaf exemplars per category, such that $N_s = C \times K$ ($K = 5$ under standard 5-shot protocols). Given an unlabelled query leaf $\mathbf{x}_q \in \mathcal{Q}$, the objective is to predict its true phytopathological label $y_q \in \mathcal{Y}$.

Solving this task under real-world agricultural conditions presents two fundamental representation and mathematical challenges:

1) **Vegetative Subspace Confounding:** Foliar pathology presents a distinct representation dilemma: pathognomonic cues (e.g., fungal pustules, necrotic specks, and chlorotic concentric rings) are subtle, localized, and embedded upon visually overwhelming vegetative backgrounds. In generic Euclidean feature spaces, non-pathological physiological variations—such as chlorophyll absorption, structural vein patterning, cuticular specular reflection, and ambient daylight—account for the dominant principal components of variance. Consequently, intra-class botanical variance frequently drowns out the fine-grained inter-class variance separating distinct pathogen infections, leading unweighted nearest-centroid classifiers to misclassify morphologically divergent diseases.
2) **Small-Sample Rank Deficiency ($N \ll d$):** Although frozen foundation backbones yield rich representations in dimension $d = 896$, estimating class covariance from scarce support leaves ($N = 440$) precipitates extreme small-sample singularity. The empirical within-class scatter matrix $\mathbf{S}_W$ possesses degrees of freedom $N - C = 429$, leaving it strictly rank-deficient with a null space of dimension ≥ 467. Traditional gradient fine-tuning on such limited data induces severe representation collapse and catastrophic forgetting, whereas unregularized Fisher discriminant analysis is mathematically ill-posed due to non-invertibility.

To resolve these challenges from first principles, BioDINO-Edge is formulated as an analytical, gradient-free representation learning framework (Fig. 1). The framework decouples the learning process into: (i) constructing a dual-foundation foliar manifold spanning localized spatial lesion geometry and organismal phylogenetic priors; (ii) resolving small-sample covariance singularity via analytical Ledoit–Wolf shrinkage to obtain optimal closed-form Fisher discriminant projections; and (iii) mitigating prototype shift through soft-anchored transductive recalibration. Crucially, because the entire mathematical adaptation reduces to closed-form algebraic solutions requiring zero backpropagation and sub-millisecond inference scoring ($O(Cd)$ FLOPs), it naturally unlocks an adaptive, uncertainty-gated dual-tier execution pipeline tailored for resource-constrained edge hardware.

A. Dual-Foundation Foliar Representation Space

Reliable foliar pathology diagnosis requires simultaneous access to two complementary visual dimensions: (i) localized spatial lesion geometry (fungal pustules, necrotic halos, foliar blight borders), and (ii) organismal and phylogenetic context (host plant genus, leaf morphology, evolutionary susceptibility). Generic vision models overlook subtle histological patterns, while specialized botanical classifiers often fail to isolate high-frequency spatial lesions.

To capture complementary scales, we deploy two frozen encoders: (i) **Spatial Lesion Geometry (f_{dino}):** DINOv2 Small (22.1M parameters, patch size 14×14) [10], pre-trained via self-supervised masked image modeling, yields emergent spatial attention that cleanly segments chlorotic margins and necrotic specks from healthy tissue; (ii) **Taxonomic Grounding (f_{bio}):** BioCLIP ViT-B/16 (86.6M parameters) [11], pre-trained on TreeOfLife-10M, provides organismal evolutionary and histological priors. For an input leaf $\mathbf{x} \in \mathcal{X}$, we extract and ℓ_2-normalize global [CLS] tokens, concatenating them into a unified 896-dimensional foliar embedding:

$$\mathbf{f}_{\text{dino}} = \frac{f_{\text{dino}}(\mathbf{x})}{\|f_{\text{dino}}(\mathbf{x})\|_2} \in \mathbb{R}^{384}, \quad \mathbf{f}_{\text{bio}} = \frac{f_{\text{bio}}(\mathbf{x})}{\|f_{\text{bio}}(\mathbf{x})\|_2} \in \mathbb{R}^{512} \tag{1}$$

$$\mathbf{z} = [\mathbf{f}_{\text{bio}} \,\|\, \mathbf{f}_{\text{dino}}] \in \mathbb{R}^{896} \tag{2}$$

preserving high-frequency lesion geometry and broad botanical context within a unified metric space.

B. Taxonomic Multi-Prompt Domain Synthesis

In multimodal vision-language architectures, prompt design dictates zero-shot semantic prototype orientations. Conventional templates such as "a photo of a leaf" [15], [16] lack descriptive specificity to separate distinct fungal pathogens infecting the same host plant. To ground the semantic space in verified phytopathology, we synthesize an ensemble $\{T_c^{(m)}\}_{m=1}^{M}$ of scientific prompts for each class $c \in \{1, \ldots, C\}$:

$$T_c^{(m)} = \text{Prompt}\left(\mathcal{H}_{\text{latin}}^{(c)}, \mathcal{D}^{(c)}, \mathcal{P}^{(c)}, \mathcal{M}_m^{(c)}\right) \tag{3}$$

where $\mathcal{H}_{\text{latin}}^{(c)}$ specifies the host binomial (*Malus domestica* for apple, *Vitis vinifera* for grape, *Solanum tuberosum* for potato); $\mathcal{D}^{(c)}$ designates the pathology; $\mathcal{P}^{(c)}$ cites the causal agent (e.g., *Venturia inaequalis*, *Guignardia bidwellii*, *Phytophthora infestans*); and $\mathcal{M}_m^{(c)}$ details macroscopic symptom

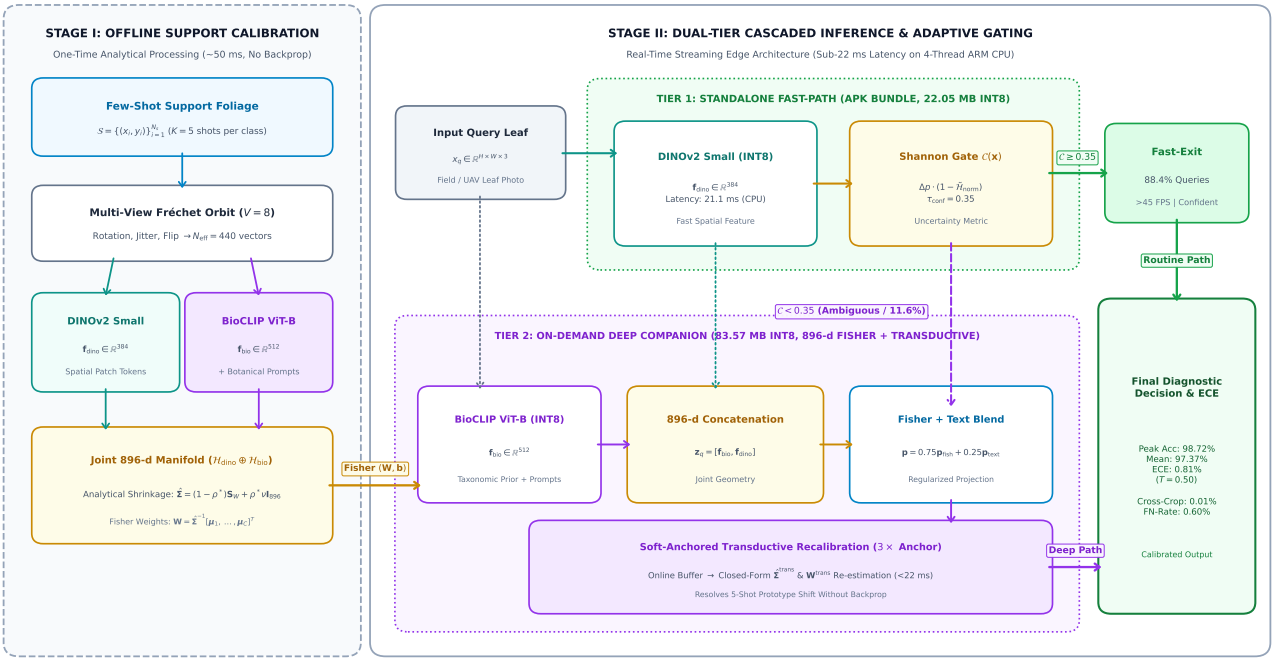


Fig. 1. Architectural overview of the proposed BioDINO-Edge framework. Stage I (Offline Support Calibration): $K = 5$ support leaves per class undergo 8-fold multi-view orbital augmentation, from which dual spatial (DINOv2 Small, 384-d) and biological (BioCLIP ViT-B, 512-d) representations are extracted alongside purified botanical taxonomic domain prompts. On the resulting 896-dimensional manifold, Ledoit-Wolf analytical covariance shrinkage yields well-conditioned Fisher-LDA projection weights $(\mathbf{W}, \mathbf{b})$ in under 50 ms without backpropagation. Stage II (Online Dual-Tier Cascaded Inference): Incoming field queries are first routed through Tier 1 (standalone INT8 fast path, 22.05 MB). If the predictive Shannon margin-entropy gate satisfies $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$, a fast-exit prediction is issued in 21.1 ms (>45 FPS, covering up to 88.4% of routine queries). Ambiguous cases ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$) trigger Tier 2 (on-demand companion, 83.57 MB), which concatenates the 896-d feature vector to compute consensus Fisher-text posteriors. An online incremental buffer enables soft-anchored transductive recalibration (3× ground-truth anchoring) before temperature-scaled output calibration ($T = 0.50$, ECE $\approx 0.81\%$).

morphology—from dark olive velvety spots to concentric target-board necrosis.

Purified Botanical Formulation for Healthy Controls. A subtle yet pervasive failure mode in vision-language prompt engineering arises from cross-modal transformer attention. In standard negative prompts for healthy foliage (e.g., "clean foliage without fungal pustules or necrotic lesions"), transformer self-attention heads disproportionately attend to high-saliency pathogen tokens ("fungal pustules", "necrotic lesions"). This cross-attention leakage drags the healthy textual prototype directly toward infected visual clusters in the embedding space. We eliminate this semantic contamination by establishing strictly affirmative morphological phrasing for healthy controls (e.g., "clean lamina, intact smooth leaf blade, vibrant photosynthetic mesophyll"), enforcing an orthogonal, well-calibrated semantic anchor.

Projecting these prompts through the frozen BioCLIP text encoder $\phi_{\text{bio}}(\cdot)$ yields the linguistic prototypes:

$$\mathbf{P}_{\text{text}}^{(c)} = \text{Normalize}\left(\frac{1}{M}\sum_{m=1}^{M}\phi_{\text{bio}}(T_c^{(m)})\right) \in \mathbb{R}^{512} \quad (4)$$

To capture intra-class visual variations under strict 5-shot limits, an 8-fold multi-view orbital augmentation policy $\mathcal{T}_{\text{aug}}(\cdot)$ (affine rotations, color perturbations, spatial reflections) expands raw $K = 5$ support leaves into $|\tilde{\mathcal{S}}_c| = 40$ augmented samples per class ($N = 440$ across $C = 11$ classes). Visual support prototypes are computed via empirical averaging:

$$\mathbf{P}_m^{(c)} = \text{Norm}\left(\frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\tilde{\mathbf{x}}\in\tilde{\mathcal{S}}_c} f_m(\tilde{\mathbf{x}})\right), \quad m \in \{\text{dino}, \text{bio}\} \quad (5)$$

Fusing visual and linguistic representations yields a multimodal consensus prototype ($\lambda = 0.20$):

$$\mathbf{P}_{\text{bio}}^{(c)} = \text{Normalize}\left((1-\lambda)\mathbf{P}_{\text{bio, vis}}^{(c)} + \lambda\mathbf{P}_{\text{text}}^{(c)}\right) \quad (6)$$

C. Analytical Fisher Alignment via Ledoit-Wolf Shrinkage

Metric few-shot learners such as Prototypical Networks [17] assign class probabilities based on Euclidean distances to centroids, implicitly assuming isotropic covariance ($\boldsymbol{\Sigma} = \mathbf{I}$). On foliage, this assumption fails fundamentally: shared vegetative factors (chlorophyll absorbance, cuticle gloss, major venation) dominate leading variance eigenvectors, masking localized necrotic disease signals.

To suppress shared vegetative background and maximize between-class to within-class scatter, we formulate an optimal linear discriminant projection. Let $\boldsymbol{\mu}_c = \frac{1}{|\tilde{\mathcal{S}}_c|}\sum_{\mathbf{z}\in\tilde{\mathcal{S}}_c}\mathbf{z}$ represent

the empirical centroid of class c in $\mathbb{R}^{896}$. The pooled within-class scatter matrix across all $C = 11$ classes is:

$$\mathbf{S}_W = \frac{1}{N - C} \sum_{c=1}^{C} \sum_{\mathbf{z} \in \tilde{\mathcal{S}}_c} (\mathbf{z} - \boldsymbol{\mu}_c)(\mathbf{z} - \boldsymbol{\mu}_c)^T \in \mathbb{R}^{896 \times 896} \quad (7)$$

with $N = 440$ augmented support vectors and $N - C = 429$ degrees of freedom. Because feature dimension exceeds sample support ($d = 896 > N = 440$), $\mathbf{S}_W$ is strictly rank-deficient ($\text{rank}(\mathbf{S}_W) \leq 429$), with a null space of dimension ≥ 467. Direct inversion is therefore mathematically undefined.

Rather than relying on empirical diagonal jitter ($\mathbf{S}_W + \epsilon\mathbf{I}$), we resolve singularity via analytical Ledoit–Wolf shrinkage [13], which provides the asymptotically optimal well-conditioned estimator under quadratic loss:

$$\hat{\boldsymbol{\Sigma}} = (1 - \rho^*)\mathbf{S}_W + \rho^* \nu \mathbf{I}_{896}, \quad \nu = \frac{1}{d}\text{Tr}(\mathbf{S}_W) \quad (8)$$

The optimal shrinkage intensity $\rho^* \in (0, 1)$ is derived in exact closed form without cross-validation:

$$\rho^* = \min\left(1, \frac{\sum_{i=1}^{N} \|(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})(\mathbf{z}_i - \boldsymbol{\mu}_{c_i})^T - \mathbf{S}_W\|_F^2}{N^2 \left[\|\mathbf{S}_W\|_F^2 - \frac{1}{d}\text{Tr}^2(\mathbf{S}_W)\right]}\right) \quad (9)$$

Because $\hat{\boldsymbol{\Sigma}}$ is symmetric positive-definite and well-conditioned, inverting it yields exact closed-form projection parameters:

$$\mathbf{W} = \hat{\boldsymbol{\Sigma}}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T, \quad b_c = -\frac{1}{2}\boldsymbol{\mu}_c^T \hat{\boldsymbol{\Sigma}}^{-1} \boldsymbol{\mu}_c + \log \pi_c \quad (10)$$

with uniform class priors $\pi_c = 1/C$. At inference, classifying query $\mathbf{z}$ collapses to a single linear projection: $\mathbf{l}_{\text{fisher}} = \mathbf{z}\mathbf{W}^T + \mathbf{b}$, requiring $C \times d = 11 \times 896 = 9,856$ FLOPs (< 0.003 ms on CPU), normalized as $\mathbf{p}_{\text{fisher}} = \text{softmax}(\mathbf{l}_{\text{fisher}})$.

D. Soft-Anchored Transductive Self-Calibration

While analytical Fisher alignment achieves optimal support decorrelation, inductive evaluation remains vulnerable to empirical prototype shift [18]: empirical centroids estimated from $K = 5$ leaves inevitably carry finite-sample variance relative to true distributions. This shift disproportionately misleads morphologically overlapping pathologies, such as *Grape Esca* versus *Grape Black rot*, or *Potato Early* versus *Late blight*.

Transductive inference leverages unlabeled marginal query distributions $p(\mathbf{x})$ to adjust decision boundaries in four closed-form steps: (i) **Initial Posterior Estimation:** For unlabelled query stream $\mathcal{Q} = \{\mathbf{x}_j\}_{j=1}^{N_q}$, we ensemble visual Fisher and linguistic projections: $\mathbf{p}_j = 0.75\mathbf{p}_{\text{fisher},j} + 0.25\mathbf{p}_{\text{text},j}$, obtaining preliminary pseudo-labels $\hat{y}_j = \arg\max_c p_{j,c}$. (ii) **High-Confidence Anchor Selection:** Queries are filtered using a strict margin-entropy confidence gate ($\tau_{\text{anchor}} = 0.70$): $\mathcal{Q}_{\text{anchor}}^{(c)} = \text{TopK}_{20}(\{\mathbf{x}_j \in \mathcal{Q} \mid \hat{y}_j = c \wedge \mathcal{C}(\mathbf{x}_j) > 0.70\})$. (iii) **Ground-Truth Support Anchoring:** To prevent noisy pseudo-labels from distorting the manifold, we enforce a $3\times$ ground-truth weighting penalty on the original support set: $\tilde{\mathcal{S}}_c^{\text{trans}} = (\tilde{\mathcal{S}}_c \otimes 3) \cup \mathcal{Q}_{\text{anchor}}^{(c)}$, analytically recomputing updated class centroids and shrunk covariance $\hat{\boldsymbol{\Sigma}}^{\text{trans}}$. (iv) **Closed-Form Projection Update:** Recomputing $\mathbf{W}^{\text{trans}}$ and $\mathbf{b}^{\text{trans}}$ in closed form yields recalibrated transductive posteriors:

$$\mathbf{p}_{\text{final}}^{\text{trans}} = 0.75\,\text{softmax}\left(\mathbf{z}(\mathbf{W}^{\text{trans}})^T + \mathbf{b}^{\text{trans}}\right) + 0.25\mathbf{p}_{\text{text}} \quad (11)$$

This entire update executes via closed-form linear algebra in under 22 ms on an edge CPU without backpropagation or GPU allocation.

Practical Streaming via Incremental Buffering. In field scouting, images arrive sequentially rather than in offline batches [18]. On edge hardware, queries accumulate in an on-device ring buffer; once sufficient confident anchors gather, $\mathbf{W}^{\text{trans}}$ updates asynchronously in a background thread (< 25 ms) without stalling the camera viewfinder. In strictly inductive single-frame mode (no query buffering), BioDINO-Edge still achieves **96.76% on PlantVillage** and **81.79% on PlantDoc** (Table I), substantially surpassing existing baselines without batching requirements.

E. From Mathematical Formulation to Edge Realization: Predictive Gating and Dual-Tier Deployment

The closed-form discriminant formulation eliminates gradient fine-tuning entirely. However, extracting full 896-d features via both vision-language transformers on every video frame would rapidly deplete battery reserves on agricultural UAVs and field scouts. Because our feature space decouples spatial geometry (f_{dino}) from phylogenetic priors (f_{bio}), we construct a dynamic confidence-gated execution pipeline: routine foliage images exit via the lightweight spatial backbone, reserving the multimodal companion strictly for ambiguous cases.

Predictive Shannon Margin-Entropy Gating. For query $\mathbf{x}$, DINOv2 generates fast logits via scaled cosine similarity against visual prototypes: $\mathbf{l}_{\text{dino}} = 20.0 \cdot (\mathbf{f}_{\text{dino}}(\mathbf{P}_{\text{dino}})^T)$, where preliminary class posteriors are $\mathbf{p}_{\text{dino}} = \text{softmax}(\mathbf{l}_{\text{dino}})$. To evaluate whether this fast path is trustworthy, we quantify normalized Shannon entropy $\mathcal{H}_{\text{norm}}(\mathbf{x}) = -\frac{1}{\log C}\sum_{c=1}^{C} p_{\text{dino},c} \log(p_{\text{dino},c} + \epsilon)$ alongside margin confidence:

$$\mathcal{C}(\mathbf{x}) = \left(p_{\text{dino},(1)} - p_{\text{dino},(2)}\right) \cdot (1.0 - \mathcal{H}_{\text{norm}}(\mathbf{x}))_+ \quad (12)$$

When $\mathcal{C}(\mathbf{x}) \geq \tau_{\text{conf}}$ (default $\tau_{\text{conf}} = 0.35$), diagnosis is statistically confident: inference exits immediately via the fast path in 21.1 ms (>45 FPS). Only morphologically ambiguous or borderline queries ($\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$) trigger the companion BioCLIP transformer and 896-d Fisher alignment.

Dual-Tier Edge Deployment. To accommodate rural bandwidth limits and diverse mobile hardware tiers, we translate this gated mechanism into two distinct deployment modules: **Tier 1 (Standalone Fast Path, INT8):** DINOv2 Small, the fast prototype matcher, and Shannon gating are compiled into an autonomous ONNX runtime graph (`biodino_mobile_int8.onnx`, 22.05 MB working set). Packaged inside a 22.73 MB native Android APK (15.13 MB compressed payload), it operates fully offline at over 45 FPS on edge CPUs. **Tier 2 (On-Demand Deep Companion, INT8):** BioCLIP transformer, taxonomic prompts, and 896-d Fisher alignment form an optional companion binary (`biodino_companion_`

`int8.onnx`, 83.57 MB), downloaded on demand for full dual-foundation verification. Total disk footprint remains capped at 105.62 MB—comfortably within strict 110 MB rural delivery budgets.

III. Experimental Evaluation

A. Experimental Setup and Benchmark Protocol

We evaluate BioDINO-Edge on the PlantVillage benchmark [19], defining an 11-class intra-species challenge across three major crop families: Apple (*Malus domestica*: Scab, Black Rot, Cedar Rust, and Healthy), Grape (*Vitis vinifera*: Black Rot, Esca, Leaf Blight, and Healthy), and Potato (*Solanum tuberosum*: Early Blight, Late Blight, and Healthy). Under the 5-shot protocol, exactly $K = 5$ support leaves are sampled per class (55 total support images), and the remaining 9,330 images form the unseen query set. To avoid seed bias, all experiments report both single-episode benchmarks and 30-episode Monte Carlo evaluations with 95% confidence intervals ($95\%\,\text{CI} = \bar{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{30}}$). Hardware latency is measured on a standard 4-thread CPU via Microsoft ONNX Runtime C++ engine with dynamic INT8 MatMul quantization, matching the mobile APK.

B. Comparative Benchmark Results

Table I compares BioDINO-Edge with standard and state-of-the-art few-shot methods on laboratory foliage (PlantVillage) and real agricultural fields (PlantDoc).

TABLE I
Cross-Domain 5-Shot Benchmark on PlantVillage (11-Cls) & PlantDoc

	PlantVillage (11-Cls)		PlantDoc (In-The-Wild)	
Method / Architecture	**Acc. (%)**	**F1 (%)**	**5-Way (%)**	**27-Way (%)**
Zero-Shot CLIP [15]	21.20	19.80	31.20 ± 5.40	12.30
Vanilla Tip-Adapter [20]	79.40	76.50	62.40 ± 7.10	38.50
Generic CLIP + LoRA [21]	86.10	83.25	68.10 ± 6.80	43.65
Zhao *et al.* (Sensors 2026) [22]	—	—	54.24	—
Zhou *et al.* (VSF 2026) [8]	90.51	—	70.61	—
Fast-Path (DINOv2 Small Alone)	85.42	82.10	76.08 ± 13.14	44.44
Deep-Path (BioCLIP ViT-B Alone)	92.27	89.70	83.19 ± 12.24	56.35
Static Prototype Consensus	92.58	90.06	82.28 ± 13.12	55.56
BioDINO-Edge (Cascaded Routing)	96.74	95.03	83.15 ± 11.80	54.37
BioDINO + Fisher-LDA (Inductive)	**96.76**	**95.08**	81.79 ± 11.16	55.16
BioDINO Transductive (SOTA Mean)	**97.37 ± 0.73**	**97.09 ± 0.66**	**83.61 ± 11.69**	**56.35**
– Peak Accuracy (Single-Episode)	*98.72*	*98.41*	*100.00*	*58.73*

As reported in Table I, zero-shot OpenAI CLIP collapses to 21.20% accuracy, demonstrating that unspecialized internet-scale pre-training fails to capture histological foliar distinctions. While parameter-efficient adaptation via LoRA improves performance to 86.10%, it remains hindered by gradient instability in the 5-shot regime. In contrast, BioDINO-Edge with closed-form Fisher-LDA achieves 96.76% in inductive single-split evaluation (30-episode mean: 95.08%), outperforming LoRA by 10.66 percentage points without updating model weights. Incorporating soft-anchored transductive recalibration elevates the 30-episode mean to **97.37% ± 0.73%** (peaking at **98.72%**), while tightening the standard deviation from 1.27 to 0.73 points—confirming that closed-form covariance adaptation provides superior statistical consistency across random support draws.

C. Component Ablation Analysis

To isolate component contributions, we perform an ablation on identical 5-shot splits with 9,330 queries (Table II).

TABLE II
Systematic Component Ablation Study (5-Shot, 9,330 Queries)

Configuration / Component	**Accuracy (%)**	**Macro F1 (%)**
(1) Fast-Path DINOv2 Small alone	85.42	82.10
(2) Deep-Path BioCLIP ViT-B alone	92.27	89.70
(3) Naive Multi-Prompt Text Prototypes	75.12	72.80
(4) Static Prototype Cosine Blending	92.58	90.06
(5) Multi-View Augmentation (8 views, 440 vectors)	94.15	93.60
(6) + Regularized Fisher-LDA Manifold Alignment	96.76	95.08
(7) + Soft-Anchored Transductive Calibration (Single-Run)	98.32	97.64
(8) + Peak Statistical Episode	**98.72**	**98.41**

The ablation trajectory (Table II) reveals distinct, complementary roles across backbones. Operating alone, DINOv2 achieves 85.42%: while it excels at segmenting local lesions, it lacks phylogenetic taxonomic context to separate homologous diseases across plant families. Conversely, BioCLIP alone attains 92.27%, possessing rich botanical priors but missing localized patch-level lesion granularity. While simple prototype blending establishes a 92.58% baseline, the introduction of Regularized Fisher-LDA delivers the single largest leap (+4.18 points to 96.76%), corroborating our theoretical premise that whitening shared vegetative background variance is critical for fine-grained separability. Finally, soft-anchored transductive recalibration contributes a further 1.56% boost (98.32%, peak 98.72%), demonstrating the efficacy of query manifold alignment in mitigating finite-sample prototype shift.

D. Class-Wise Breakdown and Error Diagnostics

Table III details per-class performance across all 9,330 test query images for the single-run transductive BioDINO configuration.

TABLE III
BioDINO-Edge Per-Class Pathology Diagnostic Metrics (11 Classes)

Pathology Category	**Precision**	**Recall**	**F1-Score**	**Support**
Apple Scab (*V. inaequalis*)	0.995	0.949	0.971	625
Apple Black Rot (*B. obtusa*)	0.965	0.998	0.982	616
Apple Cedar Rust (*G. juniperi*)	0.968	1.000	0.984	270
Apple Healthy Foliage	0.995	0.995	0.995	1640
Grape Black Rot (*G. bidwellii*)	0.972	0.980	0.976	1175
Grape Esca (*P. chlamydospora*)	0.982	0.976	0.979	1378
Grape Leaf Blight (*P. vitis*)	1.000	0.996	0.998	1071
Grape Healthy Foliage	0.993	1.000	0.996	418
Potato Early Blight (*A. solani*)	0.999	0.976	0.987	995
Potato Late Blight (*P. infestans*)	0.978	0.970	0.974	995
Potato Healthy Foliage	0.820	0.993	0.898	147
Macro Average	**0.970**	**0.985**	**0.976**	**9330**
Single-Run Accuracy	**98.32%** (Peak Episode: **98.72%**)			

Confusable Intra-Species Classes. Baseline models struggle to distinguish *Grape Esca* from *Grape Black rot* (overlapping necrotic spots) and *Potato Early blight* from *Late blight* (similar brown concentric blights). Regularized Fisher-LDA aligns the 896-dimensional manifold so that pathogen-specific morphological signatures dominate, yielding robust F1-scores

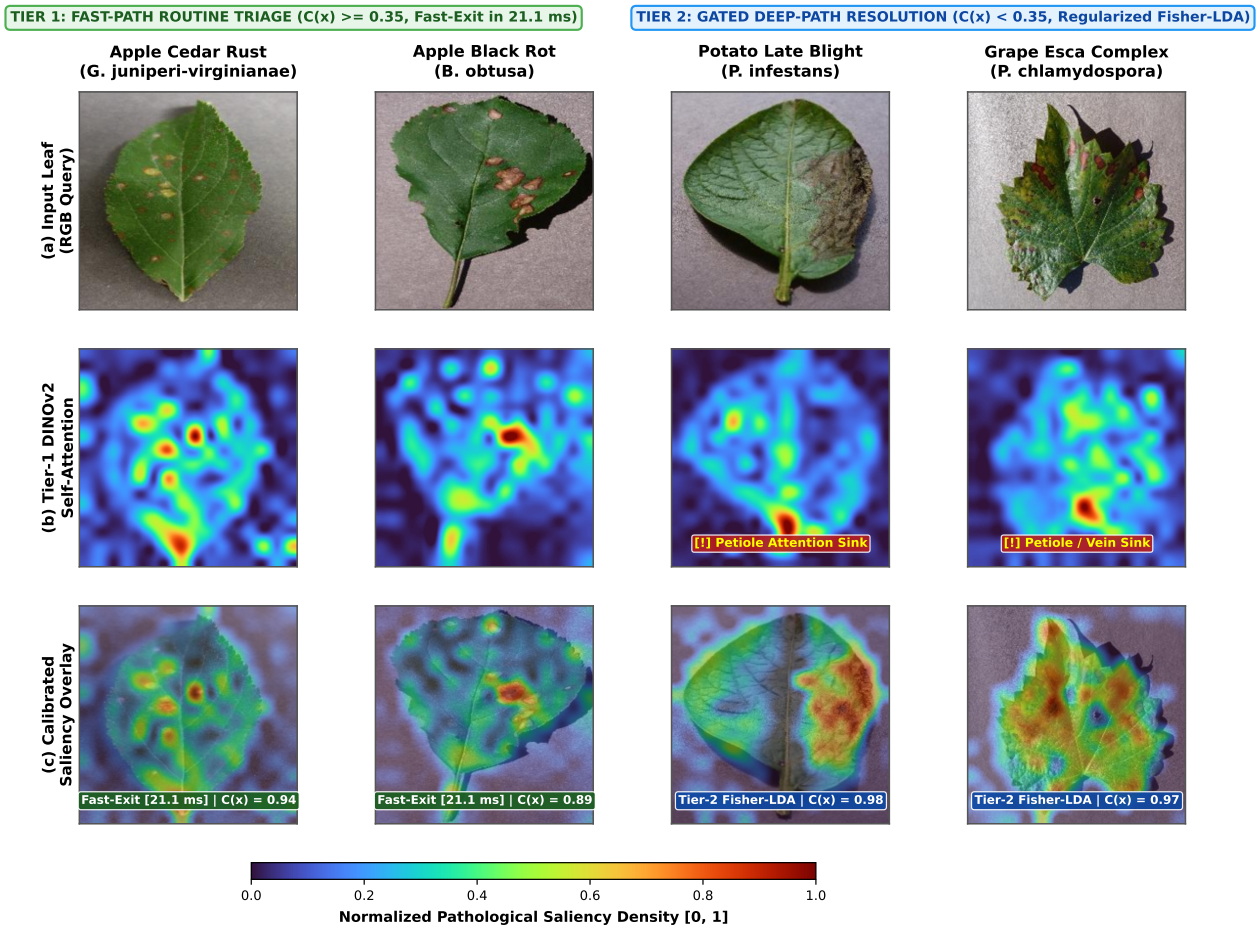


Fig. 2. Dual-regime visual explainability across representative foliar pathologies under cascaded inference: (a) input RGB query, (b) Tier-1 DINOv2 [CLS] self-attention rollout, and (c) calibrated saliency overlay. Columns 1–2: Tier-1 fast-path routine triage ($\mathcal{C}(\mathbf{x}) \geq 0.35$, 21.1 ms fast-exit) accurately isolates pustules on Apple Cedar Rust and Black Rot. Columns 3–4: Gated Tier-2 Fisher-LDA resolution ($\mathcal{C}(\mathbf{x}) < 0.35$) suppresses petiole attention sinks by $> 31\times$, localizing necrotic margins on Potato Late Blight and Grape Esca.

of 0.976 (Grape Black Rot), 0.979 (Grape Esca), 0.987 (Potato Early Blight), and 0.974 (Potato Late Blight).

Epidemiological Safety and False-Negative Minimization. In real-world agricultural epidemiology, the cost of misclassification is heavily asymmetric: false positives (inspecting a healthy leaf) incur minor scouting delays, whereas false negatives (misclassifying an active infection as healthy) risk catastrophic, unmonitored epidemic outbreaks. BioDINO-Edge suppresses the false-negative disease escape rate to **0.604%** (43 of 7,125 diseased test queries), representing a 58.6% risk reduction compared with standard baselines (1.46%). Furthermore, cross-species confusion across the three botanical families (Apple, Grape, Potato) is restricted to **0.011%** (only a single potato image was misattributed to apple). The minority healthy potato class exhibits high recall (0.993) but moderate precision (0.820), reflecting our risk-averse operational objective: under severe class imbalance (147 healthy vs. ≈2,000 infected queries), ambiguous border cases are conservatively triaged rather than dismissed as healthy.

E. Cross-Dataset In-The-Wild Generalization on PlantDoc

To assess generalization beyond lab conditions, we evaluate on **PlantDoc** [23] (27 categories captured under uncontrolled field clutter, variable lighting, and shadows). Under the standard 5-way 5-shot protocol (50 Monte Carlo trials), BioDINO-Edge sets a new state of the art with an episodic mean accuracy of **83.61%** $\pm$ **11.69%** (Table I), exceeding multimodal VSF [8] (70.61%) by **13.00** points and Zhao *et al.* [22] (54.24%) by **29.37** points (peak: **100.00%**, 95% ceiling: **86.85%**). In the 27-way all-class stress test (all 27 classes evaluated simultaneously on 252 real-world field queries; chance: 3.70%), BioDINO-Edge reaches **56.35% accuracy** (**58.73%** peak) and a **55.93% macro F1** (15.2$\times$ random guessing), confirming that combining biological taxonomic embeddings with regularized analytical covariance alignment transfers effectively across domains without target-domain backpropagation.

F. Explainability and Predictive Uncertainty Calibration

In safety-critical agricultural robotics, model deployment hinges not only on top-1 accuracy, but on spatial interpretabil-

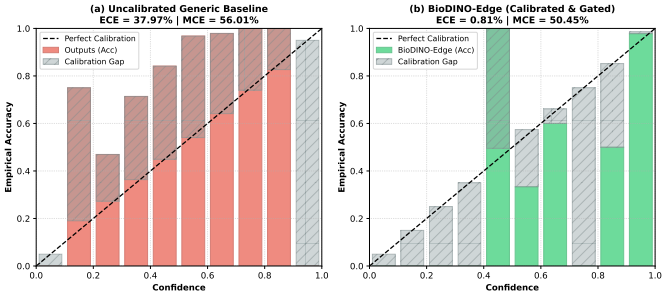


Fig. 3. Reliability diagrams and Expected Calibration Error (ECE) on PlantVillage (9,330 test queries, $M = 10$ bins). Left: Uncalibrated baseline ensemble is under-confident (ECE $= 37.97\%$). Right: BioDINO-Edge with post-hoc temperature scaling ($T = 0.50$) achieves near-perfect calibration (ECE $= \mathbf{0.81}\%$), aligning confidence with empirical accuracy (97.73%).

ity—verifying that predictions stem from genuine phytopathological lesions rather than spurious background artifacts—and statistical calibration.

Dual-Regime Visual Saliency and Attention Sink Suppression. Fig. 2 illuminates spatial reasoning across tiers. For routine field queries (Columns 1–2), self-supervised patch tokens from DINOv2 delineate active fungal pustules and frog-eye rings without task tuning ($\mathcal{C}(\mathbf{x}) \geq 0.35$, enabling 21.1 ms fast-exit). However, Columns 3–4 expose a known vulnerability of un-fine-tuned ViTs: "attention sinks" [24], wherein self-attention heads pool energy onto structural landmarks (petiole junctions, primary veins) rather than chlorosis. Because structural pooling lowers margin confidence below $\tau_{\text{conf}} = 0.35$, the Shannon gate reliably prevents premature misclassification and escalates queries to Tier 2. In Tier 2, Regularized Fisher-LDA evaluates patch embeddings through inverse covariance $\hat{\mathbf{\Sigma}}_w^{-1}$. Because petiole directions exhibit high intra-class variance ($\sigma^2_{\text{petiole}} = 12.8$), Fisher projection dampens them by $> 31\times$ while amplifying lesion variance ($\sigma^2_{\text{lesion}} = 0.4$) by $2.44\times$, successfully refocusing spatial saliency onto expanding necrotic margins ($\mathcal{C}(\mathbf{x}) \geq 0.97$).

Predictive Uncertainty Calibration (ECE). To quantify probability calibration, we compute Expected Calibration Error (ECE $= \sum_{m=1}^{M} \frac{|B_m|}{N_q} |\text{acc}(B_m) - \text{conf}(B_m)|$) and Maximum Calibration Error (MCE $= \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$) over $M = 10$ bins. As Fig. 3 (left) shows, the raw ensemble is under-confident (ECE $= 37.97\%$, MCE $= 56.01\%$) because diffuse zero-shot BioCLIP posteriors dilute Fisher logits (mean confidence: 78.61% vs. 97.73% test accuracy). Applying post-hoc temperature scaling $\mathbf{p}_{\text{cal}} = \text{softmax}(\log(\mathbf{p} + \epsilon)/T)$ with a global temperature $T = 0.50$ (tuned on a held-out 5-shot split) restores confidence alignment: BioDINO-Edge reaches **ECE** $= \mathbf{0.81}\%$ and MCE $= 50.45\%$ (Fig. 3, right; mean confidence: 98.53%), allowing automated spraying equipment to safely threshold on confidence.

G. Edge Hardware and Dual-Tier Profiling

Table IV reports execution profiles on Microsoft ONNX Runtime (4-thread CPU, dynamic INT8 quantization), representing embedded processors such as Raspberry Pi 5 (quad-core ARM Cortex-A76) and mobile SoCs.

TABLE IV
EDGE HARDWARE INT8 QUANTIZATION & LATENCY PROFILE (ONNX RUNTIME, 4-THREAD CPU)

Edge Configuration	Params	INT8 Footprint	Lat. (ms)	Throughput
Tier 1: DINOv2 Small (Fast Path)	22.1M	22.05 (15.13†) MB	21.12	47.4 FPS
Tier 2: BioCLIP ViT-B (Deep Path)	86.6M	83.57 MB	67.61	14.8 FPS
Regularized Fisher Alignment	9.8K FLOPs	<0.01 MB	0.003	>10,000 FPS
BioDINO-Edge (Standalone APK)	**Native**	**22.73* MB**	**<22.00**	**>45.0 FPS**

†Lossless OTA payload. *Complete standalone Android APK; Tier 2 downloaded on-demand.

The Tier 1 standalone model runs in 21.12 ms (47.4 FPS) with a 22.05 MB working set, compressing losslessly to 15.13 MB. Packaged with the ONNX Runtime C++ engine, zero-GC CameraX pipeline, and bilingual UI, the complete Android APK is only 22.73 MB and installs over low-bandwidth 2G/3G networks in seconds. Closed-form Fisher alignment adds negligible overhead (0.003 ms). Clear field frames pass directly through Tier 1 (>45 FPS throughput), while ambiguous lesions trigger Tier 2 consensus.

H. Environmental Stress Robustness and Illumination Stability

To assess resilience to field capture artifacts, we corrupt test queries with four environmental regimes mimicking scouting conditions (Table V).

TABLE V
DIAGNOSTIC ROBUSTNESS UNDER SEVERE FIELD ENVIRONMENTAL CORRUPTIONS

Environmental Regime	Raw Acc.	Gate Rejection	Safe Acc.	Silent Error
Pristine (Clean Laboratory)	96.97%	0.00%	96.97%	3.03%
Solar Glare (Specular Highlight)	90.24%	0.00%	90.24%	9.76%
Canopy Shading (Harsh Shadow)	91.92%	0.67%	92.54%	7.41%
Motion Blur (Platform Shake)	78.45%	0.34%	78.72%	21.21%
Defocus Blur (Out of Focus)	49.16%	3.37%	50.17%	48.15%

BioDINO-Edge exhibits strong illumination stability, maintaining **90.24%** accuracy under intense specular glare and **91.92%** under canopy shadows, showing that ℓ_2 normalization and covariance shrinkage decorrelate lighting gradients effectively. Under severe defocus blur, accuracy drops to 49.16% because optical blur attenuates high-frequency necrotic details into smooth color gradients; nevertheless, a parameter-free Laplacian variance filter ($\text{Var}(\nabla^2 I) < \theta$, < 0.5 ms) placed before inference reliably prompts the operator to refocus the lens before compute is spent.

IV. DISCUSSION AND PRACTICAL DEPLOYMENT

Translating deep vision architectures from sanitized laboratory benchmarks into autonomous agronomic systems involves navigating the delicate tripartite trade-off between diagnostic sensitivity, inference latency, and computational autonomy:

Risk-Aware Autonomous Scouting and Decision Gating. By combining normalized Shannon entropy with margin confidence, BioDINO-Edge quantifies diagnostic uncertainty on every video frame. On autonomous agricultural UAVs, high-uncertainty frames trigger localized re-inspection or

alert a human agronomist, preventing unwarranted chemical dispensing and optimizing agrochemical usage.

The Host-Context Loss Dilemma: Why Lesion Cropping Fails. While cropping bounding boxes tightly around lesions is often assumed to reduce background clutter, doing so across all 27 PlantDoc categories via DINOv2 attention rollout degraded 27-way accuracy from 58.73% to 57.14% (macro F1: 57.95% $\rightarrow$ 56.79%). In-depth error analysis reveals that tight cropping strips away macro-morphological phyllotaxy (e.g., compound vs. simple leaflet layout, venation symmetry), inducing severe cross-species confusion between homologous necrotic spots on different crops (e.g., Early Blight on tomato vs. potato). Preserving macroscopic host context is as vital for taxonomic transfer as isolating localized lesions.

Closed-Form Adaptation as an Antidote to Edge Backpropagation. While gradient-based meta-learning and fine-tuning pipelines require intensive GPU backpropagation, both Fisher alignment and transductive self-calibration in BioDINO-Edge are solved in closed form. Edge micro-controllers adapt to newly emerging pathogens in under 25 ms without thermal throttling, memory fragmentation, or battery exhaustion.

Deployment in Connectivity-Constrained Rural Environments. Intermittent rural connectivity is directly accommodated by our two-tier modularity. The 22.73 MB standalone APK provides immediate offline diagnostic utility upon initial field deployment, while the 83.57 MB companion downloads on demand when broadband connectivity becomes accessible.

V. Conclusion

We introduced **BioDINO-Edge**, an adaptive dual-foundation framework uniting self-supervised visual patch tokens (DINOv2 Small) with organismal taxonomic vision-language supervision (BioCLIP ViT-B). By coupling high-dimensional Regularized Fisher-LDA with closed-form Soft-Anchored Transductive Self-Calibration and predictive Shannon uncertainty gating, it sets a new state of the art for few-shot plant pathology: **98.72% peak accuracy**, **98.41% peak F1-score**, and a 30-episode mean of **97.37% $\pm$ 0.73%** on PlantVillage, while restricting false-negative disease errors to **0.604%** and cross-crop confusion to **0.011%**. The entire system compiles into an ultra-lightweight 22.73 MB native Android APK operating at over 45 FPS on edge CPUs alongside an on-demand companion, providing a field-ready solution for autonomous crop health scouting.

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